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Data Analytics and Data Science Research Assignment 1

Section A: Data Fundamentals

1. What is a Database?

A database is an organised collection of information that is stored electronically so that it can be easily accessed, managed, updated and retrieved when needed. In simple terms, I see a database as a digital filing system where information can be stored safely and found quickly.

Databases are essential in the modern world because businesses and organisations deal with large amounts of information every day. Databases help them keep information organised, reduce the chances of losing important data, and make it easier to find and update information. They also help organisations make better decisions because they can access accurate information when they need it.

Real-world examples of where databases are used.

• **Banks:** Banks use database to store customers' personal details, account balances and transaction records. When a customer makes a payment or withdraws money, the database is updated.

b. Schools: Schools use database to store learners' information, such as names, grades, attendance records, subjects and exam results. Teachers and administrators can easily access and update this information when needed.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data Warehouse?

A data warehouse is a centralised system used to collect, store and organise large amounts of data from different sources. It is designed mainly for reporting, analysis and decision-making rather than for handling everyday transactions. In simple terms, I think of a data warehouse as a central storage area where a company can bring together information from different systems and analyse it to identify trends and patterns.

Data warehouses are important because businesses often store information in many separate systems such as sales, finance, customer service and marketing databases. A data warehouse combines this information in one place, making it easier to analyse and compare. The data is usually cleaned, organised and stored over time so that businesses can produce reliable reports and make informed decisions.

How does a data warehouse differ from a regular database?

A regular database is mainly used to store and manage current, day-to-day information and support transactions. For example, it may record a customer making a payment, an ~~employee~~ ^{employee} updating a record or a shop processing an order. Regular database are designed for fast inserts, updates and retrieval of

individual records.

A data warehouse is mainly used for analysis and reporting. It usually contains historical data collected from multiple databases and ^{other} sources. Instead of focusing on individual transactions, it allows users to examine large amounts of information, compare performance over time and identify trends.

Some key differences include:

a. Purpose: A regular database supports daily operations, while a data warehouse supports analysis and decision-making.

b. Type of data: A regular database ~~often supports~~ ^{usually contains} current operational ~~one application~~ data, while a data warehouse stores historical data from different sources.

c. Data sources: A regular database often supports one application or business process, while a data warehouse combines data from many systems.

d. Queries: Regular databases are designed for quick transactions processing, while data warehouse are designed for complex queries and reports.

e. Users: Regular database are commonly used by applications and operational staff, while data warehouse are often used by managers analysts and business intelligence teams.

3. What is a Data lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A data lake is a large storage system where an organisation can keep different types of data in one place. The data can be stored in its original form without having to organise or process it first. This can include structured data, such as tables and numbers, as well as unstructured data, such as images, videos, emails, emails and documents.

A data lake differs from a data warehouse because a data warehouse stores data that has already been organised and processed so that it can be used for reporting and analysis. A data lake, on the other hand, can store large amounts of raw data and allows the organisation to decide later how the data will be used.

When would a business prefer a data lake? A business may prefer a data lake when it needs to store large amounts of different types of data and may want to use the data for different purposes in the future. For example, a company that collects customer information, website activity, social media data, images and sales records may use a data lake to keep all this information in one place.

Data lakes are also useful for big data, advanced analytics and artificial intelligence, because

businesses can store raw data and analyse it
when needed.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A data mart is a smaller part of a data warehouse that contains information for a specific department or area of a business. For example, a company can have a data mart specifically for the sales department, containing information about customers, sales and products.

A data mart is related to a data warehouse because it usually gets its information from the larger data warehouse. The data warehouse contains information from different parts of the business, while a data mart focuses only on the information needed by a particular department. This makes it easier and faster for employees to find and analyse the information they need.

Who typically uses a Data Mart?
Data marts are usually used by specific department and their employee. For example the sales team may use a sales data mart to analyse sale performance, while the finance team may use a finance data mart to analyse expenses and financial information. Managers and analysts also use data marts to ^{create} reports and make better business decisions.

Key Benefit

One key benefit of a data mart is that makes information easier and faster to access because

employees only see the data that is relevant to their department.

Example: A sales manager can use a sales data mart to quickly check which products are selling the most without having to search through all the information stored in the company's data warehouse.

5. Explain the difference between structured, Semi-structured and unstructured data. Provide one example of each

a. Structured data

Structured data is data that is organised in a fixed format, usually in rows and columns, making it easy for a computer to store and analyse.

Example: A customer database containing names, phone numbers, addresses and dates of birth.

b. Semi-structured data

Semi-structured data does not follow a strict table format, but it has some organisation or tags that make it easier to understand and process.

Example: An XML or JSON file, such as an online customer order containing the customer name, order number and items purchased.

c. Unstructured data

Unstructured data has no fixed or organised format and is usually more difficult for computers to analyse.

Example: Photos, videos, audio recordings, social media posts or emails

Easy way to remember:

- Structured = organised in tables → Excel spreadsheet
- Semi-structured = partly organised with tags → JSON/XML
- Unstructured = no fixed format → Photos/videos/audio.

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Modeling

Data Modeling is the process of creating a visual blueprint for data structures, attributes, and relationships before building a system, aligned with standards from IBM.

Why It Matters

- Stops bloat: Prevents duplicate records that slow down software.
- Ensures quality: Sets rules so data stays clean and reliable.
- Saves money: Catches structural mistakes before writing code.
- Aligns teams: Gives engineers and business group shared map.

Common Types of data Models (Abstraction level)

- Conceptual Model: A high-level sketch. Focuses on core business objects like customers and products.
- Logical Model: A detailed map. Adds keys and rules while ignoring specific database software.
- Physical Model: A technical blueprint. Defines actual tables, data types, and performance settings for a chosen database.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

Data mining is the process of sorting through large data sets to find hidden patterns, trends and rules.

How Business Uses Data Mining

- Customer segmentation: Grouping buyers by habits to target ads.
- ~~Market basket analysis~~ ^{Fraud detection}: Flagging weird spending patterns to stop card theft.
- Market basket: Finding items bought together to plan shelf layout or cross selling.
- Demand forecasting: Predicting stock needs to cut waste.

Real-World Examples

- Netflix Recommendation: Netflix mines viewing history, search data and pause times. It uses clustering and association rules to recommend ^{shows} and decide what new content to greenlight.
- Credit Card Fraud Flags: Visa or major banks mine real-time transaction streams (location, purchase size, frequency). Anomaly detection algorithms spot unusual charges and block fraudulent cards instantly.

8. What is Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g. MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A Database Management System (DBMS) is a software application that acts as an interface between a database and its end-users or programs, allowing users to create, read update, delete, and manage structured data.

The table below outlines four popular DBMS platforms, their core characteristics, and their most suitable use cases:

DBMS Platform	Type/Model	Description	Suitable Use Case
MySQL	Relational (CRDB MS)	A widely used, open-source relational database owned by Oracle Corporation. It is highly reliable, fast, and forms the core database component of the traditional LAMP Software Stack.	E-Commerce Website & Content Management Systems (CMS): Ideal for powering platforms like WordPress, Magento, or standard online retail sites where consistent table structures (e.g. users, products, orders) and quick read speeds are essential.

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9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

ETL stands for Extract, Transform, Load. It is a data integration process that moves raw material from separate sources into a central storage place for analysis.

The Three Steps of ETL

- **Extract:** You pull raw data from source systems (database, APIs, CSV files). Data moves into a staging area (a temporary holding space) to protect live production systems.
- **Transform:** You clean and fix the data. You remove duplicates, convert units, validate fields, and shape data to match a target schema (the structural blueprint of a database table).
- **Load:** You write the ready-to-use data into a central destination like a Google Cloud data warehouse. Loads can be full refreshes or incremental (adding only new changes).

Why ETL is Important for Data Analytics

- **Improves data quality:** Clean data stops wrong business conclusions. ETL fixes missing values, typos, and bad formats.
- **Unifies Siloed Sources:** Data lives in separate CRM, billing, and web log systems. ETL combines them into one unified view.
- **Boost query speed:** Pre-structured tables let analytics dashboards load fast. Raw messy data takes too long

to query live

- Enforces governance and security: You can mask personal data (PII) during transformation before Storage, following practices noted by IBM.

10. What are common data storage formats used in analytics (e.g. CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

Analytics relies on plain text row formats for small file sharing and binary columnar/row formats for high-speed big data processing, overviewed in the Snowflake Data Formats Guide.

Format Descriptions

- CSV: Plain-text tabular file where lines are rows and commas separate columns. Treats all values as text. Lacks a built-in schema (data blueprint).
- JSON: Text-based document format using key-value pairs and nested objects. Human-readable and flexible for hierarchical structures.
- Parquet: Binary columnar file (storing data column-by-column rather than row-by-row). Includes footer metadata for fast file skipping.
- Avro: Binary row-oriented format that bundles data with a JSON schema. Built for fast serialization (turning memory objects into byte streams).

When to Use Which Format

- CSV: Use for small tables, quick local exports, or sharing with non-technical team members via spreadsheets.
- JSON: Use for web API responses, application event payloads, or flexible multi-nested logs.
- Parquet: Use for analytical SQL queries, data lakes, and BI reporting. It reads only requested columns (column pruning) and compresses heavily.

- ~~Avro: Use for analytical SQL queries, data lakes, and~~
- Avro: Use for high-throughput write-heavy streaming (such as Apache Kafka) where safe schema evolution (backward/forward compatibility) is required

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

Definition of Narrow AI vs. General AI

- **Narrow AI (Weak AI):** Systems designed and trained to perform a single, specific task or a limited set of tasks within a predefined scope. It operates under constraints and lacks general cognitive abilities or self-awareness.
- **General AI (Strong AI):** A theoretical form of machine intelligence that matches human cognitive capabilities. It can understand, learn, reason, and apply knowledge across any intellectual task or new domain just like a human.

Key Difference

- **Scope:** Narrow AI is restricted to one narrow domain, whereas General AI can adapt and function across multiple unrelated fields.
- **Existence:** Narrow AI is the only type of AI that exists and is used in the world today, while General AI remains purely hypothetical and is still under research.
- **Learning:** Narrow AI optimizes performance for its programmed task, whereas General AI can autonomously learn and generalize skills for brand-new situations.

Examples

- **Siri:** A common example of **Narrow AI** that handles voice commands and web searches of cooking, cleaning, driving and reasoning through completely novel

real-world problems without human reprogramming.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Machine learning (ML) is a branch of **Artificial Intelligence (AI)** that trains computers to learn from data and make decisions or predictions without being explicitly programmed.

Machine Learning vs. Traditional Programming
The main difference lies in how outputs and rules are created.

- **Traditional Programming:** You input Data and hard-coded Rules into a computer and the program outputs the Answers.

- **Example:** To check if a number is even, you write a specific rule ($\text{if number \% 2} == 0$).

- **Machine Learning:** You input Data and the expected Answers into an algorithm and the computer figures out the Rules or patterns on its own.

- **Example:** You feed a computer thousands of photos of cats and dogs labeled with their correct names, and the ML system figures out what features make a photo a cat or a dog.

The Three Main Types of Machine Learning
Supervised Learning

- **Definition:** The model trains on a labeled dataset, meaning every input data point comes with the correct answer attached.

- How it works: The algorithm compares its generated output to the correct label, measures the error, and adjusts itself until it predicts the correct answers accurately.

- Common tasks:

- Classification: Assigning data into distinct groups (e.g. sorting emails into "spam" or not spam)

- Regression: Predicting continuous numerical values (e.g. estimating house prices based on size and location).

Unsupervised Learning

- Definition: The model works with unlabeled data with no answer key provided.

- How it works: The algorithm explores the raw data independently to discover hidden structures, similarities, or underlying patterns.

- Common tasks

- Clustering: Grouping similar data points together (e.g. segmenting customers by purchasing habits).

- Anomaly Detection: Finding unusual outliers or fraudulent transactions.

Reinforcement Learning

- Definition: An agent learns to make a sequence of decisions through trial and error by interacting with a dynamic environment.

- How it works: The algorithm receives feedback in the form of rewards for good actions and penalties for poor actions, adjusting its strategy over time to maximise total rewards.

- Common use cases: Training self-driving cars to navigate traffic or teaching AI to play games like chess

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

Deep Learning is an advanced subset of Machine Learning that uses multi-layered artificial neural networks to automatically learn complex patterns from raw, unstructured data.

Relationships Between AI, Machine Learning, and Deep Learning

- Artificial Intelligence (AI): The broad science of mimicking human intelligence, reasoning, and problem-solving in computers.
- Machine Learning (ML): A core subset of AI that allows computers to learn from data and make predictions without explicit step-by-step programming.
- Deep Learning (DL): A specialized subset of machine learning powered by deep neural networks with multiple hidden layers. While traditional ML requires humans to manually label and select features, deep learning discovers feature automatically.

What are Neural Network?

Artificial neural networks (ANNs) are computing systems inspired by the biological structure of the human brain. They ^{serve} as the foundational backbone of deep learning algorithms.

How Neural Networks Work

- **Input Layer:** Receives raw data, such as the pixel values of an image or audio frequencies.
- **Hidden Layers:** Interconnected stacks of nodes that perform mathematical operations. Early layers catch simple features like edges, while deeper layers recognize complex shapes or meanings.
- **Output Layer:** Produces the final prediction, classification, or decision.
- **Weights and Biases:** Each connection has a weight showing its importance, and each node has a bias to help fine-tune when it activates.
- **Training and adjustment:** The network compares its guesses to reality and adjusts its weights and biases to reduce errors through a process called backpropagation.

14. What is Natural Language Processing (NLP)?
Describe what it is and provide three real-world applications of NLP that you use in everyday life.

Natural language processing (NLP) is a branch of artificial intelligence that helps computers read, write, and understand human language.

What is Natural Language Processing?

- Definition: NLP combines computer science, artificial intelligence, and linguistics to ^{bridge} ~~systems~~ the gap between human communication and digital systems.
- How it works: It breaks down spoken or written text into data points, applies rules of grammar, and uses machine learning models to interpret meaning.
- Goal: It allows ~~Machine learning models to~~ ^{process large amounts of} natural language data and respond in a way that feels natural and human.

Three Real-World Applications in Everyday Life.

- Voice-Activated Assistants: Tools like Amazon Alexa and Apple Siri use NLP to listen to voice commands, figure out what you want, and talk back or control your smart home.
- Machine Translation: Services like Google Translate use NLP to study sentence structures and automatically translate text or speech from one foreign language into another.
- Email Spam Filters: Mail programs use NLP text classification to scan incoming messages, spot

suspicious words or phishing attempts, and block junk mail from reaching your inbox

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

A Large Language Model (LLM) is an advanced artificial intelligence system trained on massive amounts of text to understand, summarize, translate, and generate human-like language. As noted by IBM, these tools change how we interact with technology by using natural conversation instead of computer code.

LLMs vs. Standard Machine Learning Models

- **Data Type:** Standard machine learning models work best with ~~standard~~ ^{structured} data like numbers, spreadsheets, and categories. LLMs specialize in unstructured data like paragraphs, articles, and conversation.
- **Task Scope:** Standard machine learning models are trained to do one specific job, such as predicting customer churn or spotting fraud. LLMs are general-purpose tools that can switch between writing code, summarizing reports, and chatting without needing new training.
- **Output Style:** Standard machine learning models output numbers, classes or probabilities. LLMs output human-readable sentences, paragraphs, or step-by-step instructions.

Two Well-known LLMs and Their Uses

- **ChatGPT:** An interactive conversational chatbot created by OpenAI that helps users write emails,

debug code, brainstorm ideas, and study complex topics

- Gemini: A multimodal AI model developed by Google that processes text, audio, and images to translate languages, analyze data, and summarize large documents.

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

Generative AI is a type of artificial intelligence that creates new content, such as text, images, music, and code, based on patterns it learns from existing.

How Generative AI Works (High Level)

- **Training on Large Datasets:** The system studies massive amounts of information, such as billions of text pages or images, using deep learning neural networks.
- **Learning Patterns:** It identifies underlying rules, structures and relationships within the data instead of just copying it.
- **Predicting Output:** When a user gives a command or prompt, the AI uses statistical probability to generate a unique, logical response piece by piece (like predicting the next word or clearing visual noise).

Three Generative AI Tools

- **ChatGPT:** An AI chatbot that uses a large language model to understand text instructions and generate human-like essays, stories, summaries, and conversational answers.
- **DALL-E:** An AI image generator that ~~helps~~ ^{create} original, realistic pictures and artistic designs from simple descriptive text prompts.
- **GitHub Copilot:** An AI coding assistant that helps programmers write software by auto-completing code snippets, explaining functions, and debugging errors directly.

inside their code editor.

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

A training dataset is a collection of information used to teach an IBM artificial intelligence model how to recognize patterns, make predictions, and complete specific tasks.

What Is a Training Dataset?

- Input-output pairs: Examples that show the AI what a question looks like and what the correct answer should be.
- The foundation: The raw material the AI studies to build its internal rules.
- The teacher: The only source of knowledge the model has before it faces the real world.

Why Quality Matters

- Sets the limit: High quality data sets the highest possible performance level for the model.
- Enables generalization: Good data helps the AI to apply what it learned to brand-new, unseen situations.
- Machine speed impact: Flaws in traditional software affect single reports, but flaws in AI compound errors instantly at machine speed. Resources on Atlassian note that data quality governs whether AI projects succeed or fail.

What Goes Wrong If Data Is Biased or Incorrect?

- Inaccurate predictions: Wrong inputs lead directly to wrong outputs, causing the AI to fail at basic tasks.
- Systematic bias: If the data favors one group of people over another, the AI will make unfair or discriminatory decisions.
- Poor generalization: The model might memorize the bad data and fail completely when tested with real-world variety.
- Hallucinations: The AI can invent false information or irrelevant conclusions because it learned distorted patterns.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

Computer vision is a branch of artificial intelligence that trains machine to extract, analyze and understand meaningful information from digital images and videos.

How Machines See and Interpret Images

Unlike humans, who rely on biological eyes and a lifetime of intuitive context, computers "see" through a grid of numbers representing pixel brightness and color values. They process visual data using a step-by-step pipeline:

- **Image Acquisition:** Cameras or sensors capture a raw visual input, converting a scene ^{into digital} contrast, or pixel values.
~~Standardizing the size and brightness.~~
- **Pre-processing:** Algorithms clean the data by removing digital noise, adjusting contrast, or standardizing the size and brightness.
- **Feature Extraction:** Deep learning models (like Convolutional Neural Networks) scan the pixels to identify foundational patterns, such as sharp edges, color gradients, and textures.
- **Object Detection and Classification:** The system groups features into complex shapes, assigns labels to objects (like a "pedestrian" or "stop sign"), and maps their exact locations.
- **Decision Making:** The software interprets the context of the identified object to trigger an automated action, such as applying brakes or flagging a medical

anomaly.

Real-World Use Cases

- **Autonomous Vehicles:** Self-driving systems use real-time computer vision to detect lane markings, identify traffic lights, track pedestrians, and navigate roads safely.
- **Medical Imaging:** Healthcare providers use vision models to scan X-rays, MRIs, and CT scans, helping doctors spot early signs of tumours, fractures, and other critical abnormalities with high precision.

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

An API (Application Programming Interface) is a set of defined rules, protocols, and definitions that allows different software applications to communicate and exchange data with one another. Essentially, it acts as a digital middleman or translator, enabling separate systems to interact seamlessly without needing to know how the other system is built internally.

Restaurant Analogy

The easiest way to understand an API is to imagine you are dining at a restaurant:

- The Customer (The client/your App): You ^{are looking} ~~sit at a table~~ at a menu of choices and want to order food. You represent a mobile app or browser trying to request information.
- The Kitchen (The server/Database): The kitchen has all the ingredients, code and systems needed to prepare your meal. However, you can't just walk into the back and start digging through their inventory yourself.
- The Waiter (The API): The waiter is the bridge between you and the kitchen. You give your order to the waiter (the API request), the waiter delivers it to the kitchen, waits for the chef to cook it, and then brings the finished dish back your table (the API response).

Why APIs Are Important in Modern Systems

APIs are the invisible backbone of modern software and data ecosystems. They are crucial for several reasons:

- Preventing "Reinventing the Wheel": Developers don't need to write every feature from scratch. If an application needs a map, they don't launch satellites - they connect to the Google Maps API. If they need to accept credit cards, they use the PayPal API.

This saves immense amounts of time and development costs.

- System Interoperability: Software is built using different programming languages and databases. APIs serve as a universal language translator, allowing completely separate systems (like an Android app and a legacy banking database) to work together seamlessly.

- Enhance Security: APIs protect core company systems. Instead of exposing an entire database to the internet, an API acts as a controlled gateway. It shares only the small, specific packets of data requested while keeping internal code and data infrastructure locked down safe.

- Automation and Efficiency: APIs allow data to sync automatically in real-time. For example, ride-sharing apps use APIs to simultaneously pull traffic data, track passenger GPS coordinates, and process payments without needing human manual setups for every transaction.

20. What is an API key? What is its purpose? How should it be stored and protected, and what could happen if it exposed publicly?

An API key (Application Programming Interface key) is a unique, randomly generated string of alphanumeric characters that acts as a digital identification card or password for an application.

Purpose of an API Key

API keys serve three primary functions for API providers:

- Authentication and Authorization: They identify the calling application and verify that it has permission to access the requested data or features

How to store and protect API keys

Because API keys grant access to services and data, they must be guarded with the same care as a master password.

- Never Handcode keys
- Use Environment variable
- Keep keys on the Backend
- Use secrets Managers
- Restrict Key Permissions

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

~~REST~~ REST API (Representational State Transfer Application Programming Interface) is an architectural style that allows different software applications to communicate with each other over the internet. It relies on standard web protocols - primarily HTTP - to transfer data between a client (like a mobile app or browser) and a server. In a REST API, everything is treated as a "resource" (such as a user, a product, or a photo), which is identified by a unique URL.

REST stands for Representational State Transfer. It is designed to be stateless, meaning every single request from the client must contain all the information needed for the server to understand and process it.

Main HTTP Methods Used in REST APIs

REST APIs standard HTTP verbs to standard database operations, commonly known as CRUD (Create, Read, Update, Delete).

HTTP Method	CRUD Operation	Description	Example
GET	Read	Retrieve data from the server without modifying anything. It is a safe and repeatable request.	Requesting a list of products or viewing a user's profile.
POST GET	Create	Submits new data to the server to create a new resource.	Signing up a new user or submitting a new blog post form.
PUT	Update	Update or completely replaces an existing resource. If the resource doesn't exist, it can sometimes create it.	Updating user's entire profile information (name, email, and password).
DELETE	Delete	Removes a specific resource from the server completely.	Deleting an account or removing a photo from a gallery.

22. What is (JavaScript Object notation?) Why is it commonly used when working with APIs and data exchange?
Write a simple example of a JSON Object

JSON stands for JavaScript Object notation, and it is a lightweight, text-based format used for storing and exchanging data.

Why JSON is commonly used for APIs and Data Exchange

- Language Independent: Although its syntax comes from JavaScript, JSON is plain text and works with almost all programming languages, including Python, Java, C++ and PHP

Lightweight: It has a clean and minimal structure without heavy tags. Unlike older formats like XML, which makes data transfer fast across web and mobile networks.

Standard for Web-APIs: Most modern APIs use JSON as the default format to send request and return responses between a server and a client application.

JSON:

"id": 101

"name": "Aminah"

"is Enrolled": true

"courses": ["Math", "Science"]

"address": {}

23. What is prompt Engineering? Why is it an important skill when working with AI like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each

Prompt engineering is the skill of writing clear and specific instructions (called prompts) that help AI tools such as ChatGPT or Claude understand exactly what you want and give you a better answer

Why is prompt engineering an important skill when working with AI?

Prompt engineering is important because:

- 1) It improves the quality of answers - A clear prompt helps AI provide more accurate and useful information
- 2) it ~~saves~~ saves time - You are less likely to keep asking the AI to correct or change its answer
- 3) It gives you more control - You can tell the AI the format, tone, length and type of answer you need.
- 4) It reduces misunderstandings - giving the AI enough context makes it easier for it to understand your actual goal.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or Javascript.

In programming and data processing, a map function is a built-in tool that transforms a collection of data by applying a specific rule or function to every single item in that collection.

Instead of updating the original list, it processes each individually and returns a brand-new collection the transformed results.

How map Works

Think of map as an automated assembly line:

- The Input: A collection of items (like a list, array, or data column).
- The operation: A function or rule you define (e.g. ~~Multiplying~~ "multiply by 2" or "convert text to uppercase").
- The Process: map takes the first item, applies the rule, moves to the second item, applies the rule, and repeats until the end.
- The Output: A brand-new collection of the exact same length, containing the newly modified data.

Use Cases in Data Processing

- Data Cleaning: Converting an entire column of messy text into lowercase format.
- Math Transformation: Converting a list of temperatures from Fahrenheit to Celsius.
- Extracting Information: Pulling just the email addresses out of a large list of user profile objects.

Code Example (Python)

python

1. Define the transformation rule

```
def calculate_tax(price):
```

```
    return price * 0.10
```

2. Create the initial collection

```
usd_prices = [10.00, 25.00, 50.00, 100.00]
```

3. Apply the map function

```
# (In Python, we wrap map() in list() to view the final  
result easily) tax_amounts = list(map(calculate_tax,  
usd_prices))
```

```
print(tax_amounts)
```

```
# Output: [1.0, 2.5, 5.0, 10.0]
```

Direct Comparison: Manual Loops vs. Map

Feature	Using a Manual Loop (for loop)	Using the map function
Code Style	Imperative (tells the computer how to step through list)	Declarative (tells the computer what transformation to make)
Readability	Requires creating empty lists and manually appending items	Cleaner, shorter, and expresses intent instantly
Performance	Can be slower in high-volume data frameworks	Highly optimized, especially (like Pandas or Spark)

25. What is Cloud computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

Cloud computing is the delivery of computing services such as servers, storage, databases, software, and networking over the Internet instead of using local computers or physical servers. It allows users and businesses to access technology resources whenever they need them and pay only for what they use.

The three main cloud service models are:

1. Infrastructure as a Service (IaaS)

2. Software as a Service (SaaS)

Example product: Microsoft 365 →

3. Platform as a Service (PaaS)

Service Model	Description	Example Product
IaaS	Provides virtual servers, storage, and networking	AWS EC2
PaaS	Provides a platform for developing applications	Google App Engine.
SaaS	Provide ready-to-use software online	Microsoft 365

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are

~~Cont~~ Version Control System

A version control system is a tool that keeps track of changes made to files or code. It helps you see what was changed and go back to an older version if needed.

Git

Git is a popular version control system. Data analysts and developers use it because it helps them track their work, collaborate with others, and avoid losing important changes.

- **Commit:** A saved record of changes you have made.
- **Branch:** A separate version of the project where you can work on new changes without affecting the main project.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

Jupyter Notebook

A Jupyter Notebook is a tool where you can write and run code, while also adding notes, explanations, tables, and graphs in the same document.

The programming language most commonly used with it is Python.

It is popular with data analysts and data scientists because they can work with data, run calculations, create graphs, and see the results immediately in one place.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

Python is a programming language used to write instructions for computers. It is very popular in data analysis and AI because it is easy to learn and has many useful libraries.

Four common Python libraries are:

1. Pandas - Used to organise, clean and analyse data
2. NumPy - Used for calculations and working with numbers and arrays.
3. Matplotlib - Used to create graphs and charts
4. Scikit-learn - Used to build machine-learning models.

29. What is Data Privacy? Why is it important?
Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to

Data Privacy

Data privacy means protecting a person's personal information and making sure it is collected and used properly. It is important because it prevents personal information from being misused, stolen, or shared without permission.

GDPR

The GDPR (General Data Protection Regulation) is a European law that protects people's personal information and gives them more control over their data. It applies in the EU countries and the wider European Economic Area (EEA). It can also apply to companies outside Europe if they offer services to or monitor people in the EU.

GDPR is there to make sure organisations handle people's personal information safely and responsibly.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

Data Ethics means using data and AI in a fair responsible and honest way.

Three examples of unethical use:

1. Sharing personal information without permission
- This violates people's privacy
2. AI discrimination - An AI system may unfairly reject people because of their race, gender or background.
3. Using AI to spread fake information - This can mislead people and cause harm

31. ~~What~~ What is Data Governance? Define it and explain the key components of a good data governance teamwork.

Data governance is the way an organisation manage and control its data to make sure it is accurate, safe and used properly.

Key components include

- Data quality - keeping data accurate and reliable
- Data Security - protecting data from unauthorised access
- Policies and rules - setting clear guidelines for using data.
- Roles and responsibilities - making sure people know ~~who~~ who manages and protects the data

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

AI bias happens when an AI system give unfair or inaccurate results because of bias in the data it was trained on

Example: If an AI hiring system is trained mostly on successful male applicants, it may unfairly prefer men over women.

Consequence: Qualified people may be rejected unfairly, leading to discrimination and unequal opportunities.

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills and tools.

Role	Main job	Key skills	Common tools
Data Analyst	Finds useful information from data	Excel, SQL, Statistics	Excel, Power BI, SQL
Data Scientist	Uses data to predict and solve problems	Python, Statistics, machine learning	Python, Pandas, Scikit learn
Data Engineer	Builds and deploys AI systems and manages systems that store and move data	SQL, databases, programming	Python, SQL, Spark
AI Engineer	Builds and deploys AI systems	Python, machine learning, AI	Python, TensorFlow, PyTorch.
Analysts understand data, Scientists predict with data, Engineers build the data systems and AI Engineers build AI solutions.			

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

Business Intelligence (BI) means using data to understand how a business is performing and make better decisions.

Two popular BI tools are:

- Power BI - Creates reports and dashboards to help business understand their data.
- Tableau - Turns data into easy-to-understand charts and visualisations.

BI helps businesses use data to make smarter decisions instead of guessing.

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The concept I found most interesting was AI bias. I was surprised that AI can make unfair decisions because of the data it learns from. It showed me how important it is to use fair and accurate data.

36. The concept I found most difficult was data governance because there are many rules and responsibilities involved.

I used Google and ChatGPT to explain it in simpler terms and help me understand it better.

37. How do you see Data and AI affecting ^{my} ~~your~~ future career or everyday life? Be specific with examples.

Data and AI can help me in my future career by making work faster and easier. For example, I can use data to understand business sales and use AI to analyse information or create reports.

In everyday life, AI can help me learn new skills, plan my work, and make better decisions.

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